Some brief notes on the $2D$ Darcy flow problem with a single fracture

Sofiane Ezzehi

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1 Problem statement

1.1 Strong forms

1.1.1 $2D$ single fracture Darcy flow problem

The $2D$ single fracture Darcy flow problem closely follows a benchmark problem presented in [1]. The problem considers a steady-state, incomressible, steady-state Darcy flow in a square domain $\Omega = (0,1)^2$ that presents a diagonal fracture that models a sudden change in the permeability field caused by a heterogeneity in the medium.

Let $\Omega = (0,1)^2$ be a square domain. Imposing mass conservation, and substituting into it Darcy's law, we obtain the following PDE,

$$\begin{aligned} -\text{div}\,(D\nabla h) &= f \quad \text{in} \quad \Omega, \\ h &= \bar{h} \quad \text{on} \quad \partial\Omega; \end{aligned} \tag{1}$$

assuming that there exists $0 < D_{\min} \leq D(x) \leq D_{\max} < \infty$ almost everywhere in Ω.

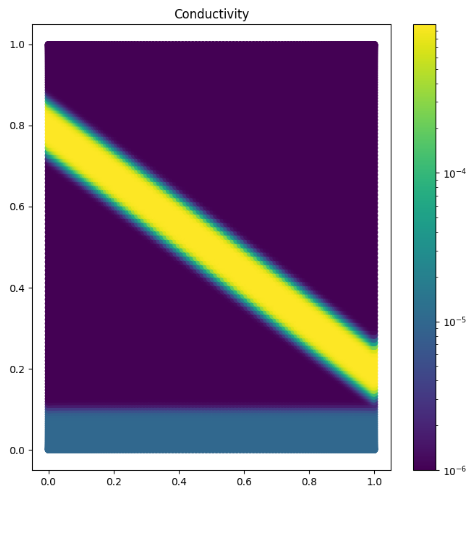

Figure 1: Permeability profile D.

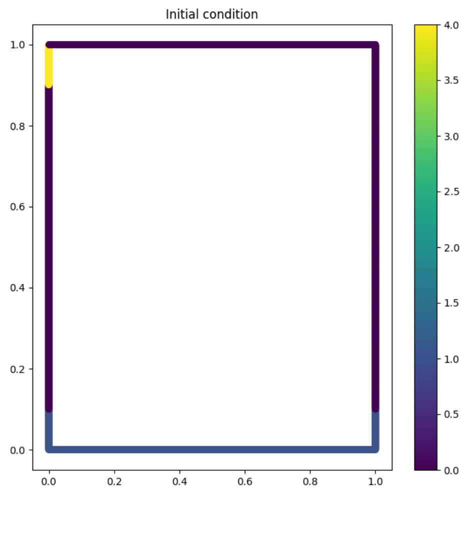

Figure 2: Boundary condition $\bar{h}$.

1.2 Derivation of the weak form

We are looking for the solution $h \in H^1(\Omega)$ of the PDE with $h = \bar{h}$ on $\partial\Omega$. This boils down to finding a solution in the affine space $H_0^1(\Omega) + h_0$ where h_0 is in $H^1(\Omega)$ and satisfies the boundary condition $h_0 = \bar{h}$ on $\partial\Omega$.

Here the space $H^1(\Omega)$ is the Sobolev space defined as,

$$H^1(\Omega) = \{u \in L^2(\Omega) : \nabla u \in (L^2(\Omega))^d\}, \tag{2}$$

and $H_0^1(\Omega)$ is the subspace of $H^1(\Omega)$ defined as,

$$H_0^1(\Omega) = \{u \in H^1(\Omega) : u = 0 \text{ on } \partial\Omega\}. \tag{3}$$

The weak form of the PDE is obtained by multiplying the PDE by a test function $v \in H_0^1(\Omega)$ and integrating over the domain Ω,

$$\int_\Omega -\mathrm{div}\,(D\nabla h)\, v\, dx = \int_\Omega f v\, dx, \tag{4}$$

where f is a source term, set to 0 in our case.
Using the identity $\mathrm{div}(v\, D\nabla h) = v\,\mathrm{div}(D\nabla h) + D\nabla h \cdot \nabla v$, followed by the divergence theorem, we obtain,

$$\int_\Omega D\nabla h \cdot \nabla v\, dx - \int_{\partial\Omega} v\, D\nabla h \cdot n\, ds = \int_\Omega f v\, dx. \tag{5}$$

Since v is a test function in $H_0^1(\Omega)$, we have $v = 0$ on $\partial\Omega$, and the boundary term vanishes. Therefore the weak form of the PDE is to find $h \in H^1(\Omega)$ such that $h = \bar{h}$ on $\partial\Omega$ and

$$\int_\Omega D\nabla h \cdot \nabla v\, dx = \int_\Omega f v\, dx, \quad \forall v \in H_0^1(\Omega). \tag{6}$$

To ensure the boundary condition $h = \bar{h}$ on $\partial\Omega$, we introduce $h_0 \in H^1(\Omega)$ such that $h_0 = \bar{h}$ on $\partial\Omega$ and set $u = h - h_0 \in H_0^1(\Omega)$. Then u satisfies

$$\int_\Omega D\nabla u \cdot \nabla v\, dx = \int_\Omega f v\, dx - \int_\Omega D\nabla h_0 \cdot \nabla v\, dx, \quad \forall v \in H_0^1(\Omega).$$

Therefore, the weak form of the PDE is to find $u \in H_0^1(\Omega)$ such that,

$$a(u, v) = l(v), \quad \forall v \in H_0^1(\Omega), \tag{7}$$

where the bilinear form a and the linear form l are defined as,

$$a(u, v) = \int_\Omega D\nabla u \cdot \nabla v\, dx, \quad l(v) = \int_\Omega f v\, dx - \int_\Omega D\nabla h_0 \cdot \nabla v\, dx. \tag{8}$$

Once we have solved for u, we can recover the solution h by adding back the function h_0 that satisfies the boundary condition, i.e.,

$$h = u + h_0.$$

1.3 Finite element discretization

To discretize the problem, we introduce a finite element space $V_h \subset H_0^1(\Omega)$, and we look for an approximate solution $u_h \in V_h$ such that,

$$a(u_h, v_h) = l(v_h), \quad \forall v_h \in V_h. \tag{9}$$

Here, we can choose V_h to be the space of piecewise linear functions on a discretization of the domain Ω into triangles (or quadrilaterals). Therefore, a basis of V_h is given by the set of hat functions $\{\phi_i\}_{i=1}^N$ associated to the nodes of the mesh. At each node i, we have $\phi_i(x_j) = \delta_{ij}$, where $\{x_j\}_{j=1}^N$ are the nodes of the mesh.

Writing u_h as a linear combination of the basis functions, we have,

$$u_h = \sum_{i=1}^N u_i \phi_i, \tag{10}$$

where u_i are the coefficients to be determined. Substituting this into the weak form, we obtain,

$$\sum_{i=1}^N u_i a(\phi_i, v_h) = l(v_h), \quad \forall v_h \in V_h. \tag{11}$$

Since this must hold for all $v_h \in V_h$, we can choose $v_h = \phi_j$ for $j = 1, \ldots, N$, which gives us a system of linear equations,

$$\sum_{i=1}^N u_i a(\phi_i, \phi_j) = l(\phi_j), \quad j = 1, \ldots, N. \tag{12}$$

This can be written in matrix form as,

$$Au = b, \tag{13}$$

where A is the stiffness matrix with entries $A_{ij} = a(\phi_i, \phi_j)$, u is the vector of unknown coefficients u_i, and b is the load vector with entries $b_j = l(\phi_j)$.

References

[1] I. Berre, W. M. Boon, B. Flemisch, A. Fumagalli, D. Gläser, E. Keilegavlen, A. Scotti, I. Stefansson, A. Tatomir, K. Brenner, S. Burbulla, P. Devloo, O. Duran, M. Favino, J. Hennicker, I.-H. Lee, K. Lipnikov, R. Masson, K. Mosthaf, M. G. C. Nestola, C.-F. Ni, K. Nikitin, P. Schädle, D. Svyatskiy, R. Yanbarisov, and P. Zulian. Verification benchmarks for single-phase flow in three-dimensional fractured porous media. *Advances in Water Resources*, 147:103759, Jan. 2021. arXiv:2002.07005 [math].